AI/NLP CASE STUDY · FINANCIAL RISK

RiskPulse

From unstructured news and social text to live risk signals that rebalance an index and stress-test a bank portfolio.

52	2	2	8/8
signals scored	text sources	modules built	tests passing

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PROBLEM AND APPROACH

Turning text into decisions

1

The gap

Markets react to headlines and tweets within minutes. Risk systems only consume numbers.

2

The engine

One NLP core reads two sources and outputs sentiment, event type and impact for every document.

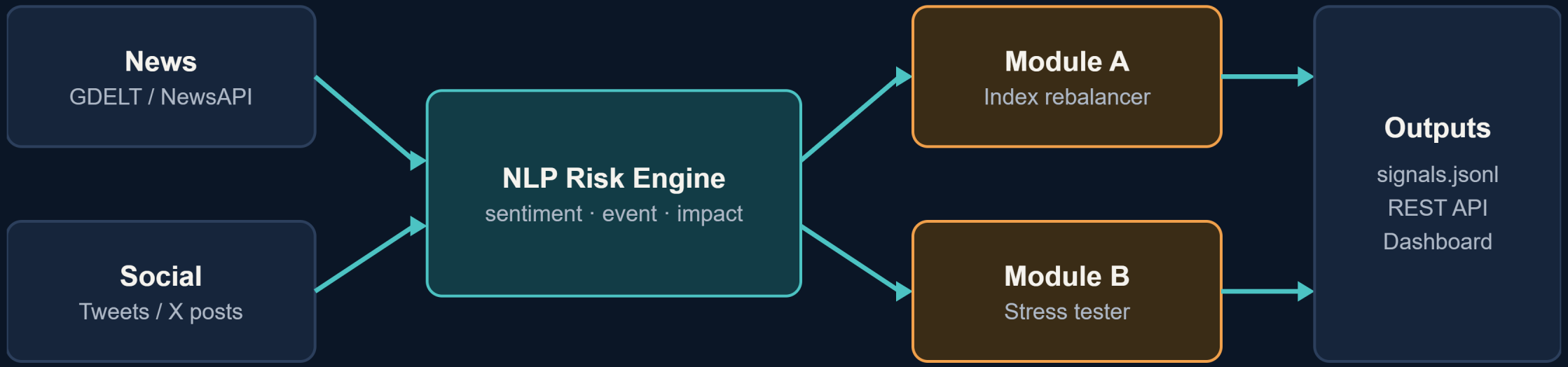
3

The proof

Both downstream modules run on the same signals: a tactical rebalancer and a strategic stress tester.

ARCHITECTURE

One engine, two downstream modules



Modules subscribe to the engine: A listens for sentiment, B listens for event type and impact.

NLP RISK ENGINE

Three structured outputs per document

SENTIMENT SCORE

-1.0 to 1.0

VADER with a finance lexicon. FinBERT can be switched in.

EVENT CLASS

7 types

Geopolitical, Macro, Credit, M&A, Product, Regulatory, Earnings.

IMPACT SCORE

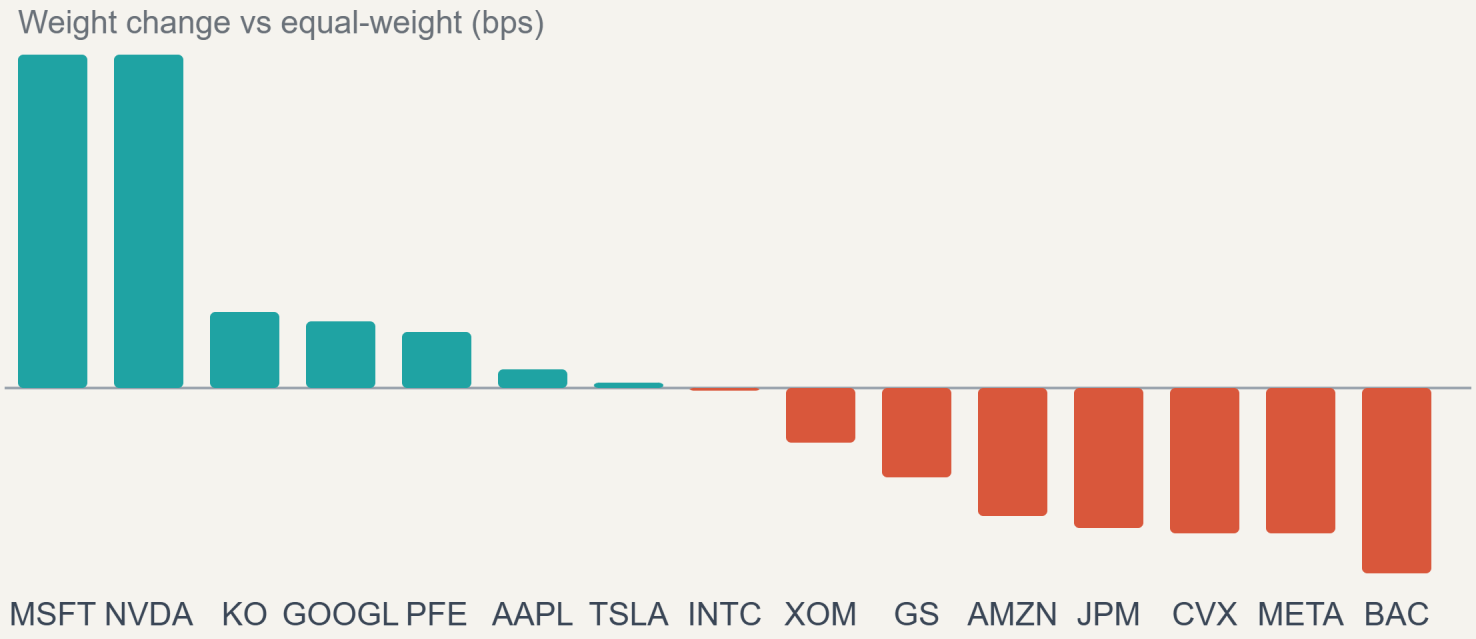
1 to 10

Event severity, intensity words, sentiment size, source weight.


```
{"tickers":["GS"], "sentiment_score":0.494, "event_type":"Earnings", "impact_score":4.5,
"confidence":0.85}
```

MODULE A · TACTICAL

Sentiment-driven index rebalancer



How it works

Smoothed sentiment times impact tilts each weight. Caps of 2% to 15% keep the index diversified.

Result

NVDA and MSFT hit the 15% cap (+833 bps). BAC falls to the 2% floor (-464 bps).

MODULE B · STRATEGIC

Event-driven portfolio stress test

Event	Equity	Rates	Spreads
Geopolitical	-10%	+50 bp	+120 bp
Macroeconomic	-7%	+200 bp	+60 bp
Credit Event	-12%	-30 bp	+250 bp
Regulatory	-4%	0	+30 bp

Triggers when impact is above 7. Shocks scale with impact on 40 positions: loans, bonds, derivatives and equity.

Geopolitical event, impact 10

-5.0%

$2,435m to $2,313m

9 stress tests fired

-3.7% to -5.7%

CHALLENGES AND NEXT STEPS

What we learned

Challenges

Noisy social text, keyword ambiguity, and illustrative shock sizes.

Mitigations

EMA smoothing, weight caps, source weights, dedupe and offline fallback.

Next steps

Fine-tuned FinBERT, streaming ingestion, calibrated scenarios, backtesting.

SOURCE CODE
github.com/ayushisehgal/riskpulse

LIVE DEMO
riskpulse-as.streamlit.app